Patron-Made Exo-Leg Mods

Crash Makerspace
16 dic 2025

2 3

Bendix Sondermann created his own bulk-versions of the components of the leg. This 3D File drop is for the full member list! These offer extra clearances and additional strength enhancements through the chunkiness of each connector print. This is a fun prototype to print out to further drive more ideas and tests for your own leg-exo systems! The same Exo-Hinges of mine will interface in the middle.

These are for educational & research purposes only, leave dangerous activities like wearing these on your body up to professionals in safe environments and configurations.

Archivos adjuntos

Descargar todo

- Compression Spring Chamber Lower Leg.STL
- Spring Retention System Max Draw.STL
- ScrewcoverLegplate.STL
- Spring Guide for Compression Spring.STL
- Upper Leg Clamp Shell.STL

Exo Hinges
6 publicaciones Ver colección

Patron-Made Exo-Leg Mods
16 dic 2025

Knee Exo Spring Mount & Guide
21 nov 2025

Metal Exo Knee
2 nov 2025

Lower Leg Clamp Shell.STL

3 comentarios

Únete a la conversación...

Bendix Sondermann · 3/mes

Sorry for the late Reply: Holyday Season, Flu and whatnow... But, there are Updates. :)For the Longer/Stronger Spring Retention Hardware ist the Same, just a n M10 Nut and an M5 Screw and Nut. The Leg Clamp Shells should be Split in a the Slicer that you use, and they use 12x M4 + Nuts for Mounting then. If Upper Leg Shell sits too Loose, thicken the Inside of it with Some Duct Tape, or Sand Both Halfs down a Bit, so they sit not flush but clamp down a little when you tighten them. The Lower Leg Shell houses the Compression Spring Chamber, That the lower two M10 Bolts run through. Here you also need at least one Nut, maybe Two, dependant on what you do with the Lower Part of the Threaded Rod that protrudes there. The Compression Spring Chamber contains a regular Airsoft Spring (mines are about M130, going by the Rating you buy them...roughly 10 Kilos.). Through this Spring goes a Huge Woodscrew that should be about 170-180mm long, and is covered with a thin TPU Tube, so the Ridges of the Screw dont interfere, and the Spring can glide and compress unobstructed in the Cylinder. Response will change dependant if you use a linear spring or a non linear one, in theory non linear should be easier to draw, But as in this Scenario the Spring is permanently under slight Compression, this Effect is most probably less pronounced. There is actually a Part missing too, and that is the TPU Eyelet that the Tip of this sharp Woodscrew screws into. The Eyelet then connects to the 1mm Dyneema Cord, which on the other End runs over the Metal Spacers of the Hinge and is directly connected to the Main Spring. The Idea is, when setting up the Leg, that you first load the Main spring and Cord that connects to the Center Armature of the Hinge to the Maximum, and then load this lighter Spring and its Cord to about its half draw (on about 25cm Rods for all Leg Connections). This takes a lot of Load of the First Cord Mounting Point (else instantly gets ripped out on everything i tried except Hard TPU with the stronger Main Spring draw). Its a Load Bypass. Also: The second spring acts as a counter-force in the starting position. This lowers the initial torque required to begin the bending motion.
As the second spring "runs over" the joint, its leverage (moment arm) changes. This can be used to neutralize the rapidly increasing resistance of the main spring, making the movement feel smoother and more consistent. Unlike a single spring (which usually feels linear), this dual-spring setup creates a custom force profile. You get a "flatter" resistance curve during the middle of the rotation. (Feels more natural)
While the movement might feel easier due to the leverage, the total potential energy in the system is higher. I had to research the Data, as it felt "weaker" at first. :D At extreme angles, the forces of both springs may eventually add up, leading to a stronger snap-back or holding force in the fully bent position. Takes some Load of your Knees, if youre kneeing down with one or two. And:
Stability at the Pivot: The opposing pull of the second spring helps stabilize the joint's axis, reducing play or "wobble" during the transition. This helped a lot with a Center Armature prone to pulverizing itself.

1

Luke · 4/mes

Is there additional hardware needed for this design?

1

Crash Makerspace AUTOR · 4/mes

The same hardware should work! The lower spring sectic
know more details on if there is a special spring down th
the cable at the bottom of the hinge still

6 publicaciones

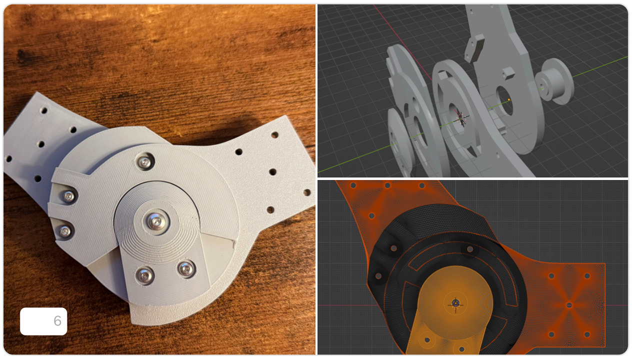
Patron-Made Exo-Leg Mods
16 dic 2025

Knee Exo Spring Mount & Guide
21 nov 2025

Metal Exo Knee
2 nov 2025

Cerrar respuestas

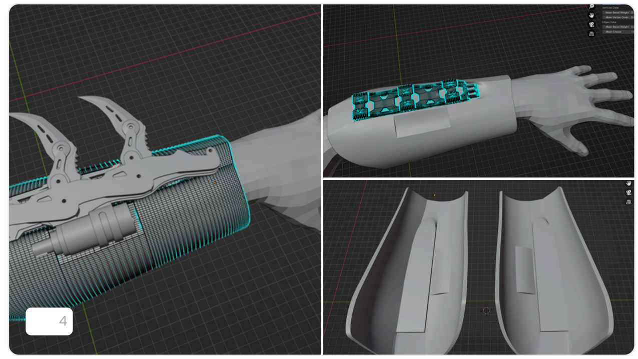
Publicaciones relacionadas >



Uniaxial Exo Hinge

The 3D model for the Exo-hinges that I designed! The multi-contact rotation limiters prevent over extension of elbows & knees in both...

12 may 2025 4 9



Vambrace Mold w/ Inserts & Mount

Vambrace Mold file with the insert for mounting the Retractable Blades to. With the Insert part the strap mounts on the blades are no longer

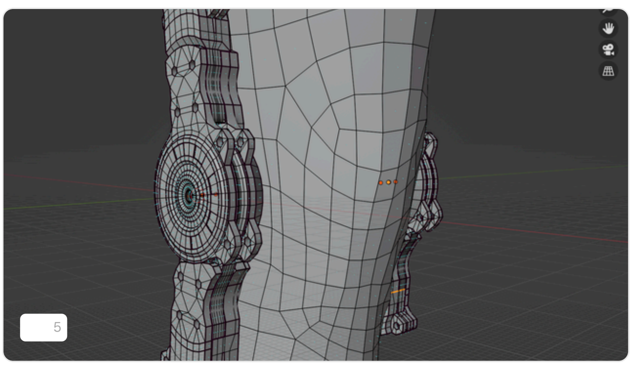
30 oct 2025 4

Sna

Buil
Gric

4 de

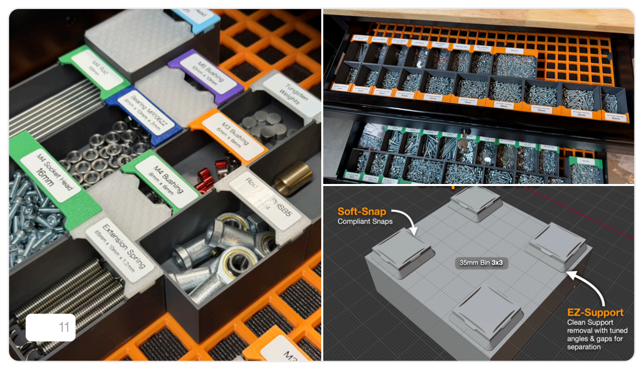
Publicaciones populares >



Knee Exo Hinge

These are the STL files for printing the Cabled Knee Hinge with tension cable track, over-extension, and threaded rod interfacing! PRINTING

28 sept 2025 6 38



Snap-Cell Organization System

Built around a 25mm cell grid (significantly higher resolution than Gridfinity's 42mm), Snap-Cell Organizer enables tighter layouts,...

4 de enero 5 9

Fut

Mal
my

2 de